MSG No. 164.86 $P\bar{3}m11'$ [Type II, trigonal]

Table 1: Wyckoff site: `1a`, site symmetry: `-3m.1'`

No.	position	mapping
1	[0, 0, 0]	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]`

Table 2: Wyckoff site: `1b`, site symmetry: `-3m.1'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]`

Table 3: Wyckoff site: `2c`, site symmetry: `3m.1'`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,3,10,11,12,13,14,15,22,23,24]`
2	$[0, 0, -z]$	`[4,5,6,7,8,9,16,17,18,19,20,21]`

Table 4: Wyckoff site: `2d`, site symmetry: `3m.1'`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[1,2,3,10,11,12,13,14,15,22,23,24]`
2	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[4,5,6,7,8,9,16,17,18,19,20,21]`

Table 5: Wyckoff site: `3e`, site symmetry: `.2/m.1'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	`[1,4,7,10,13,16,19,22]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[2,5,8,11,14,17,20,23]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[3,6,9,12,15,18,21,24]`

Table 6: Wyckoff site: `3f`, site symmetry: `.2/m.1'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[1,4,7,10,13,16,19,22]`
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,5,8,11,14,17,20,23]`
3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,6,9,12,15,18,21,24]`

Table 7: Wyckoff site: `6g`, site symmetry: `.2.1'`

No.	position	mapping
1	$[x, 0, 0]$	`[1,4,13,16]`
2	$[0, x, 0]$	`[2,5,14,17]`

continued ...

Table 7

No.	position	mapping
3	$[-x, -x, 0]$	`[3,6,15,18]`
4	$[-x, 0, 0]$	`[7,10,19,22]`
5	$[0, -x, 0]$	`[8,11,20,23]`
6	$[x, x, 0]$	`[9,12,21,24]`

Table 8: Wyckoff site: `6h`, site symmetry: `.2.1'`

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	`[1,4,13,16]`
2	$\left[0, x, \frac{1}{2}\right]$	`[2,5,14,17]`
3	$\left[-x, -x, \frac{1}{2}\right]$	`[3,6,15,18]`
4	$\left[-x, 0, \frac{1}{2}\right]$	`[7,10,19,22]`
5	$\left[0, -x, \frac{1}{2}\right]$	`[8,11,20,23]`
6	$\left[x, x, \frac{1}{2}\right]$	`[9,12,21,24]`

Table 9: Wyckoff site: `6i`, site symmetry: `.m.1'`

No.	position	mapping
1	$[x, -x, z]$	`[1,11,13,23]`
2	$[x, 2x, z]$	`[2,12,14,24]`
3	$[-2x, -x, z]$	`[3,10,15,22]`
4	$[2x, x, -z]$	`[4,9,16,21]`
5	$[-x, x, -z]$	`[5,7,17,19]`
6	$[-x, -2x, -z]$	`[6,8,18,20]`

Table 10: Wyckoff site: `12j`, site symmetry: `11'`

No.	position	mapping
1	$[x, y, z]$	`[1,13]`
2	$[-y, x-y, z]$	`[2,14]`
3	$[-x+y, -x, z]$	`[3,15]`
4	$[x-y, -y, -z]$	`[4,16]`
5	$[y, x, -z]$	`[5,17]`
6	$[-x, -x+y, -z]$	`[6,18]`
7	$[-x, -y, -z]$	`[7,19]`
8	$[y, -x+y, -z]$	`[8,20]`
9	$[x-y, x, -z]$	`[9,21]`
10	$[-x+y, y, z]$	`[10,22]`
11	$[-y, -x, z]$	`[11,23]`
12	$[x, x-y, z]$	`[12,24]`